PoKémon ML + Zoomorphology

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1 Introduction

- Here I will need to add a simple introduction to give context to this project.
- Bla bla Bla.
- This is a franchise created by **GameFreak**

Target of This project: To predict variables of new Pokémon in the next videogame of the franchise to be released in 2027.

This documentation is divided as follows: ...

2 Explanation of PoKémon features (Dataset)

The dataset used in the project will be the *PoKédex*. This encyclopedia can be thought as the animal/creature atlas of the Pokémon world. It collects information of all known species.

So far, it contains information of up to 1025 Pokémon species, adding up to 1082 different ones if one accounts for regional variants.[1]

Every Pokémon can be categorised based on its basic attributes (i.e. the dataset columns):

[1] Ex: African Elephant and Indian Elephant. Regional variants are Pokémon within a specie that were created to have different values on their features if belonging to a different Pokémon world region.

- `BST`: Base Stat Points. In general terms, it describes how "strong" a Pokémon is. It ranges from 175 to 720.
- `type(s)`: like `electric` (i.e. Pikachu), `fire`/`flying` (i.e. Charizard), etc. Total 18 different ones as unique or pairs.
- `height`: 0.1 to 20 m.
- `weight`: 0.1 to 999.9 kg.
- `color`: The main color of the Pokémon, among red, blue, yellow, green, black, brown, purple, gray, white, and pink. See this link for more information on how the main color is decided.
- `shape`: There are 14 different shapes (`upright`, `quadruped`, etc.). See this link for more information.
- `stage`: Pokémon can evolve. A Pokémon cannot evolve more than 2 times. See next section 3 for further information.
- `rarity`: Majority of Pokémon belong to the `regular` rarity. There are `legendary` ones like Articuno, Moltres, Mewtwo. And `mythical` ones, like Mew.

Game Freak, the Pokémon Company, usually provide the first three ones when a new Pokémon is announced. (For example, see these three new Pokémon for the new games to come in 2027). The rest of the `features` (`color`, `shape`, `stage` and `rarity`) are easily extracted from visuals and little experience within the franchise. `Stage` requires more experience with the franchise (i.e. one would never expect a Charizard to be the first stage of an evolution chain of two stages!, but other Pokémon, like Archaludon, could be hard to determine at sight).

3 Data Analysis

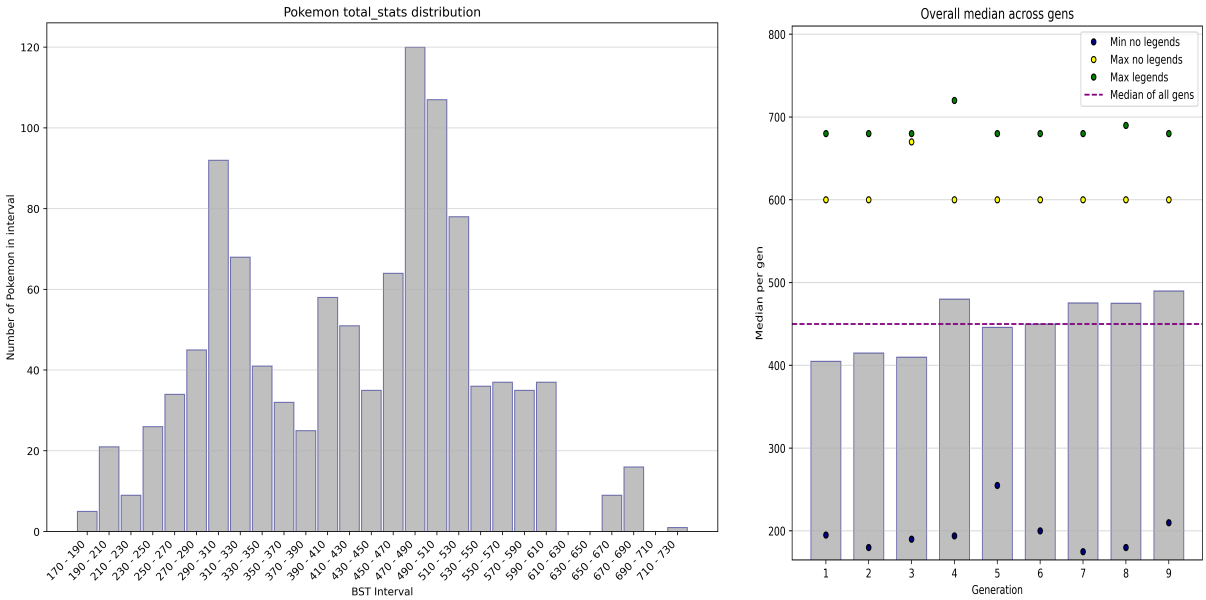
3.1 BST Distribution Across Generations

Pokémon is an old franchise which started in 1996. Therefore, not all 1082 different species were "discovered" at the same time. With each new generation of games, new Pokémon are included, increasing the quantity of these creatures. At this point, we can be interested in two main aspects:

1. How the `BST` value spreads over its whole range, i.e. the amount of creatures that share a given interval of `BST`.
2. The median `BST` for each generation.

In fact, we can see the following distributions:

The left plot 1a shows how Pokémon are distributed from the minimal value of `BST` to the maximum one in intervals of 20 units. In fact, we see that there are two main cluster intervals; (270–330) and (450–490). We will later see what these correspond to when comparing to the `stage` in the evolution chain. Morever, we can see an isolated cluster in the upper `BST` values, pointing to the unique strength of `legendary` creatures.



(a) BST Intervals

(b) Median Evolution

Figure 1: BST distribution analysis

The right plot 1b shows the median of BST for each generation that has been released in the last 30 years. This median seems to agree with the second cluster of BST in the first image. Overall, we can see that the median has been increasing over generations. This may point to the fact that there exists some power-creep (i.e. Pokémon are getting stronger and stronger designs in each generation) that could be useful when training the model.

This plot also displays the minimum and maximum value of BST for each generation (corresponding to one or more Pokémon in that generation). We can also see the overall median across all generations. It is also worth noting the behaviour of min/max values for each generation:

- The minimum value seems to be under or equal to 200 points (BST of 175, corresponding to Wishiwashi), with generations 5th and 9th having outliers (Patrat and)
- On the other hand, if we exclude `legendary` Pokémon, we see that the maximum value for BST of `non-legendary` Pokémon is quite constant across generations, except for generation 3[2]. This seems to impose an upper limit for any `non-legendary` Pokémon that has not been broken in a long time.
- For `legendary` Pokémon we see the value seems also to be limited up by 700 points. Only generation 4 crossed this barrier.[3]

[2] Game Freak designers were slacking off in that generation...

[3] Arceus, the "God" of all Pokémon, with total BST of 720 was introduced in this generation. It makes sense to set this value as the upper-limit for BST among all pokemon, even those ones to come in the future.

3.2 BST distribution by evolution chain stage and category

Let us now explore how the `BST` behaves for each evolution chain stage and category. First, it is important to explain how we are going to catalogue each stage in an evolution chain.

- `Babies2`: First stage of an evolution chain of 2 stages. (i.e. Ekans, Rattata, etc)
- `Babies3`: First stage of an evolution chain of 3 stages. (i.e. Charmander)
- `F2`: Second stage of an evolution chain of 2 stages. (i.e. Sandslash)
- `F3`: Third stage of an evolution chain of 3 stages. (i.e. Charizard)
- `Inter`: Second stage of an evolution chain of 3 stages. (i.e. Charmeleon)
- `Single`: No evolution chain. (i.e. Ditto)

Therefore, if we use this way to catalogue and split each evolution stage, we then see:

CONTINUE HERE!!!

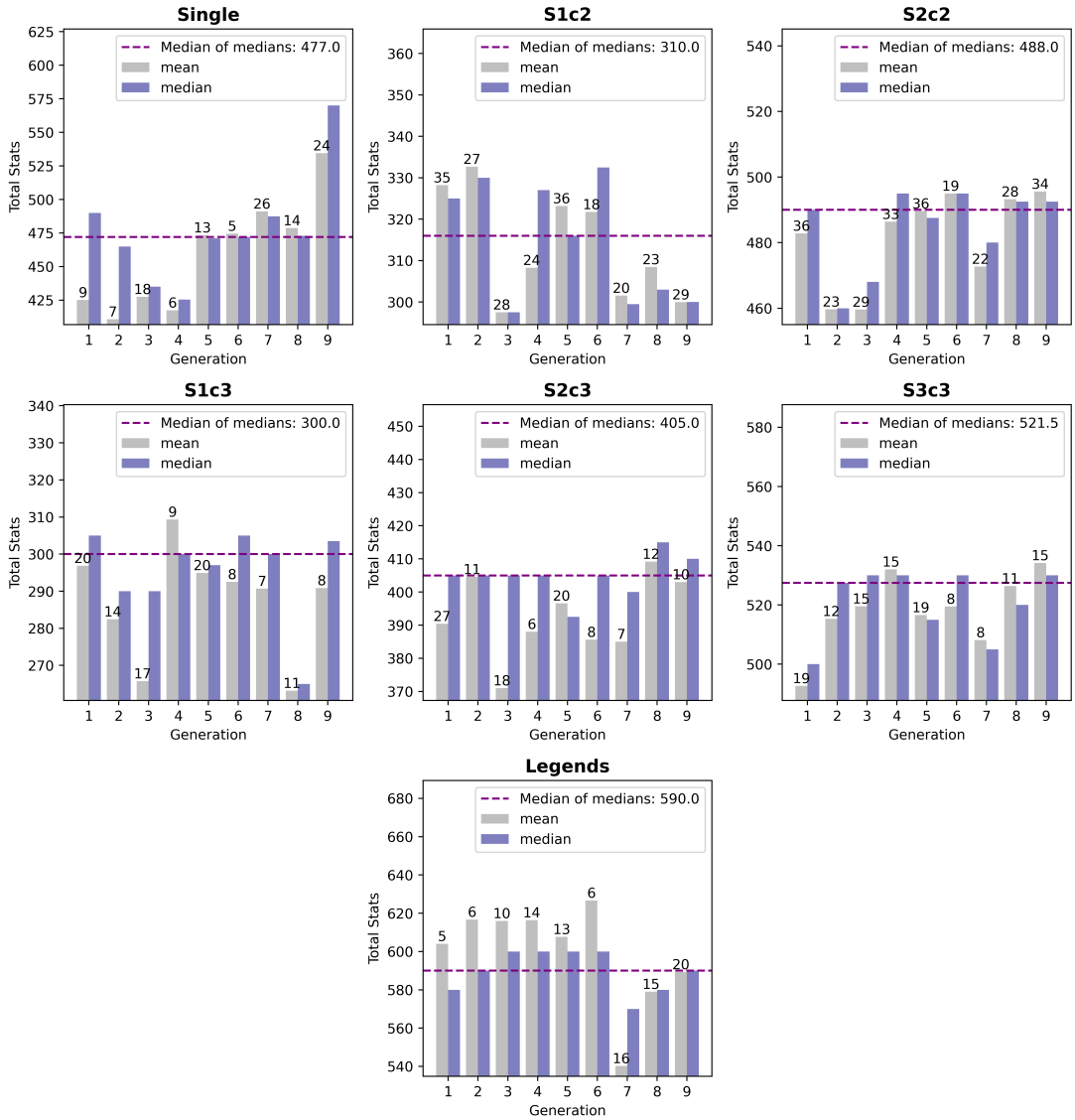


Figure 2: Distribution of mean/median across each generation for each evolution stage.

Here we can also extract juicy information:

- **F3 (final stage, 3-stage chain):** This trend seems to follow a cubic polynomial around the median.
- **Babies (pre-evo):** No clear pattern regardless of chain length.
- **Singles:** While generations 1 and 2 had powerful single evolution pokemon, it decreased in the two following ones and kept a steady growth from Gen 5.
- **Inter (stage 2 of 3):** This seems to nicely adjust to the overall median across generation. However, there is a slight favorability from Gen 7+.

- **Legendaries:** Gen 7 dip due to Cosmog line, which is a `legendary` pokemon with an evolution chain of 3 stages.

In fact, it could be illuminating if we try to fit different polynomials to 2 clear patterns in F3 and Singles (from third generation onwards)

F3 Mean/Median (cubic fit):

F3 polynomial fit

The $R^2 > 0.7$ for the median indicates strong ongoing trend! This can result useful for **feature engineering**!

Singles Linear Fit (Gen 3+):

Singles linear fit

As we suspected, the median trend for this stage is quite strong also, $R^2 \approx 0.7$. This can also be important to keep in mind when **feature engineer** new attributes.

4 Some ideas for Feature Engineering

- While the idea of projecting all stages to a single mock one could be tempting, I am affraid it would be better to add the type-ordering effects map for each stage rather than a generic fake projection. It would be more realistic at the end of the day.
- I guess the map should be like two extra columns one for primary and other secondary. If there is data, then project a normalised value wrt to the maximum value of that stage?
- Question here is what to do with single types or types without info? Should I input to 0?